

Full forms and meanings

Alphabetical reference to course terminology. Meanings state the relevant context; lesson links open the detailed daily PDFs.

[A-C](#) · [D-F](#) · [G-L](#) · [M-P](#) · [Q-S](#) · [T-Z](#)

SDC has distinct logic and timing meanings. CDC here is controllability don't-care. Liberty is a format name; tCQ, tSU and tH are symbols. Operation names and conventional abbreviations are identified by context.

A-C

ASIC	Application-Specific Integrated Circuit A chip designed for a particular application.	Lesson 02
AST	Abstract Syntax Tree Hierarchical representation of parsed source syntax.	Lesson 15
ATE	Automatic Test Equipment Applies and measures manufactured-device test patterns.	Lesson 08
ATPG	Automatic Test Pattern Generation Derives patterns to detect modeled manufacturing faults; fault-model coverage differs from functional coverage.	Lesson 08
BCP	Boolean Constraint Propagation Repeated deduction from unit clauses.	Lesson 23
BDD	Binary Decision Diagram Graph representing recursive Boolean decisions.	Lesson 21
BEOL	Back End of Line Fabrication of interconnect and related structures after the transistor/device stages; not the entire physical-design workflow.	Lesson 01
BIST	Built-In Self-Test On-chip pattern generation and response evaluation for a defined test target.	Lesson 08
BMC	Bounded Model Checking Checks unfolded behavior through a specified cycle bound.	Lesson 24
CCS	Composite Current Source More detailed current-based library model.	Lesson 26
CDC	Controllability Don't-Care Impossible internal input combination caused by network dependencies.	Lesson 19
CEC	Combinational Equivalence Checking Compares corresponding combinational functions.	Lesson 25
CMOS	Complementary Metal-Oxide-Semiconductor Uses complementary n-channel and p-channel devices; the inverter illustrates pull-down and pull-up behavior.	Lesson 01
CNF	Conjunctive Normal Form AND of clauses, each an OR of literals.	Lesson 23

CPU	Central Processing Unit Executes an instruction set; an architecture may allocate some behavior to software running on it.	<u>Lesson 04</u>
CTS	Clock Tree Synthesis Builds a clock-distribution network.	<u>Lesson 07</u>
D-F		
DC	Don't-Care Behavior permitted to vary under justified specification conditions.	<u>Lesson 17</u>
DEF	Design Exchange Format Exchanges a design instance's physical implementation data, such as placement, nets and routing.	<u>Lesson 07</u>
DFT	Design For Test Structures that improve manufactured-device testability.	<u>Lesson 08</u>
DPLL	Davis-Putnam-Logemann-Loveland SAT search through decisions, implications and backtracking.	<u>Lesson 23</u>
DRC	Design Rule Checking Checks geometric manufacturing rules.	<u>Lesson 08</u>
DUT	Design Under Test Implementation exercised by a verification environment.	<u>Lesson 13</u>
ECO	Engineering Change Order Controlled implementation change followed by the relevant functional, timing and physical rechecks.	<u>Lesson 07</u>
ECSM	Effective Current Source Model Current-source model introduced alongside NLDM and CCS.	<u>Lesson 26</u>
EDA	Electronic Design Automation Tools for design, implementation and verification.	<u>Lesson 05</u>
EPI	Essential Prime Implicant Prime covering a required minterm uniquely among primes.	<u>Lesson 17</u>
ERC	Electrical Rule Checking Checks electrical connection and usage rules defined by the technology and rule deck.	<u>Lesson 08</u>
FEOL	Front End of Line Fabrication stages that form device structures, before most interconnect processing.	<u>Lesson 01</u>
FPGA	Field-Programmable Gate Array Configurable logic and routing programmed after manufacture.	<u>Lesson 02</u>
FSM	Finite State Machine State, transition and output model for sequential behavior.	<u>Lesson 20</u>

G-L

GBA	Graph-Based Analysis Propagates timing bounds through graph vertices.	<u>Lesson 29</u>
GDSII	Graphic Data System II Layout exchange format used for geometric mask data.	<u>Lesson 09</u>
HDL	Hardware Description Language Models concurrent hardware behavior and structure.	<u>Lesson 11</u>
HLS	High-Level Synthesis Schedules and binds behavioral operations into hardware.	<u>Lesson 14</u>
IC	Integrated Circuit Devices and interconnect integrated on a substrate.	<u>Lesson 01</u>
IP	Intellectual Property Reusable design block and supporting integration information.	<u>Lesson 04</u>
LEF	Library Exchange Format Abstract physical information for cells and routing.	<u>Lesson 26</u>
Liberty	Liberty timing-library format Cell functions, timing, units and power models in .lib files.	<u>Lesson 26</u>
LVS	Layout Versus Schematic Checks extracted layout connectivity against a design reference.	<u>Lesson 08</u>

M-P

MMMC	Multi-Mode Multi-Corner Checks justified combinations of mode and timing conditions.	<u>Lesson 30</u>
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor Gate voltage controls channel conduction; complementary device types form CMOS logic.	<u>Lesson 01</u>
NAND	NOT-AND Boolean complement of AND; its output is zero only when every input is one.	<u>Lesson 17</u>
NBA	Nonblocking Assignment Schedules a later left-side update after evaluating the right side.	<u>Lesson 13</u>
NLDM	Nonlinear Delay Model Lookup models of delay and transition versus slew and load.	<u>Lesson 26</u>
NMOS	N-channel Metal-Oxide-Semiconductor N-channel MOS device; the CMOS inverter uses it for the pull-down path.	<u>Lesson 01</u>
NOR	NOT-OR Boolean complement of OR; its output is one only when every input is zero.	<u>Lesson 17</u>
OCV	On-Chip Variation Models local timing variation using appropriate early/late effects.	<u>Lesson 30</u>
ODC	Observability Don't-Care Internal variation masked at all relevant observed outputs.	<u>Lesson 19</u>

OPC	Optical Proximity Correction Adjusts mask geometry to compensate for predictable lithographic printing distortion.	Lesson 09
PBA	Path-Based Analysis Recomputes selected paths with path-consistent quantities.	Lesson 29
PDK	Process Design Kit Process models, layers and design rules.	Lesson 01
PDN	Power Delivery Network Connected supply/ground structures whose impedance affects local voltage, timing and reliability.	Lesson 07
PI	Prime Implicant A legal product cube that cannot be further expanded.	Lesson 17
PMOS	P-channel Metal-Oxide-Semiconductor P-channel MOS device; the CMOS inverter uses it for the pull-up path.	Lesson 01
POS	Product of Sums AND of OR terms in Boolean notation.	Lesson 17
PPA	Power, Performance, Area Interacting implementation quality measures.	Lesson 02
PVT	Process, Voltage, Temperature Operating/characterization conditions of timing models.	Lesson 30
Q-S		
RC	Resistance-Capacitance Interconnect resistance and capacitance contribute to delay, transition and loading.	Lesson 29
ROBDD	Reduced Ordered Binary Decision Diagram Canonical for one fixed variable order, with reduction rules applied.	Lesson 21
RPI	Redundant Prime Implicant In the essentials-first classification, a nonessential prime whose ON points are already covered by selected essentials.	A-13 cover selection
RTL	Register Transfer Level Describes transfers and transformations between state elements.	Lesson 15
SAT	Boolean Satisfiability Existence of an assignment satisfying a Boolean formula.	Lesson 23
SDC (logic)	Satisfiability Don't-Care Impossible relation between local inputs and their computed output.	Lesson 19
SDC (timing)	Synopsys Design Constraints Tcl-based design/timing requirements.	Lesson 32
SEC	Sequential Equivalence Checking Compares behavior over legal sequences and related initial states.	Lesson 25

SoC	System on Chip Integrates major system functions and reusable blocks on one integrated circuit.	<u>Lesson 04</u>
SOP	Sum of Products OR of AND terms in Boolean notation.	<u>Lesson 17</u>
SPEF	Standard Parasitic Exchange Format Extracted interconnect resistance and capacitance information.	<u>Lesson 29</u>
SPI	Selective Prime Implicant Remaining nonessential prime that can cover a still-uncovered ON point; select a sufficient residual cover.	<u>A-14 worked chart</u>
STA	Static Timing Analysis Analyzes timing paths with models and constraints.	<u>Lesson 28</u>
T-Z		
TB	Testbench Stimulus, checking and supporting simulation logic.	<u>Lesson 13</u>
Tcl	Tool Command Language Command language used by many EDA interfaces.	<u>Lesson 10</u>
tCQ	Clock-to-Q delay Delay from the launch clock pin to changing Q.	<u>Lesson 28</u>
tH	Hold time Required data stability after a capture edge.	<u>Lesson 28</u>
tSU	Setup time Required data stability before a capture edge.	<u>Lesson 28</u>
UNSAT	Unsatisfiable No satisfying assignment under the encoded conditions.	<u>Lesson 23</u>
VCD	Value Change Dump Timestamped signal changes for waveform inspection.	<u>Lesson 13</u>
VLSI	Very Large Scale Integration Integration scale motivating a staged design flow.	<u>Lesson 03</u>
XOR	Exclusive OR For two inputs, the output is one when they differ; multiple-input XOR represents odd parity.	<u>Lesson 17</u>

Reference trail

Meanings are concise study summaries. Each lesson link retains the lecture frames, timestamps and supporting sources. ATPG/BIST have later treatment in Weeks 9-10; FEOL/BEOL have further treatment in Week 10.

[NPTEL official course outline](#) · [Course lectures](#) · [Source register](#) · [Flow guide](#)